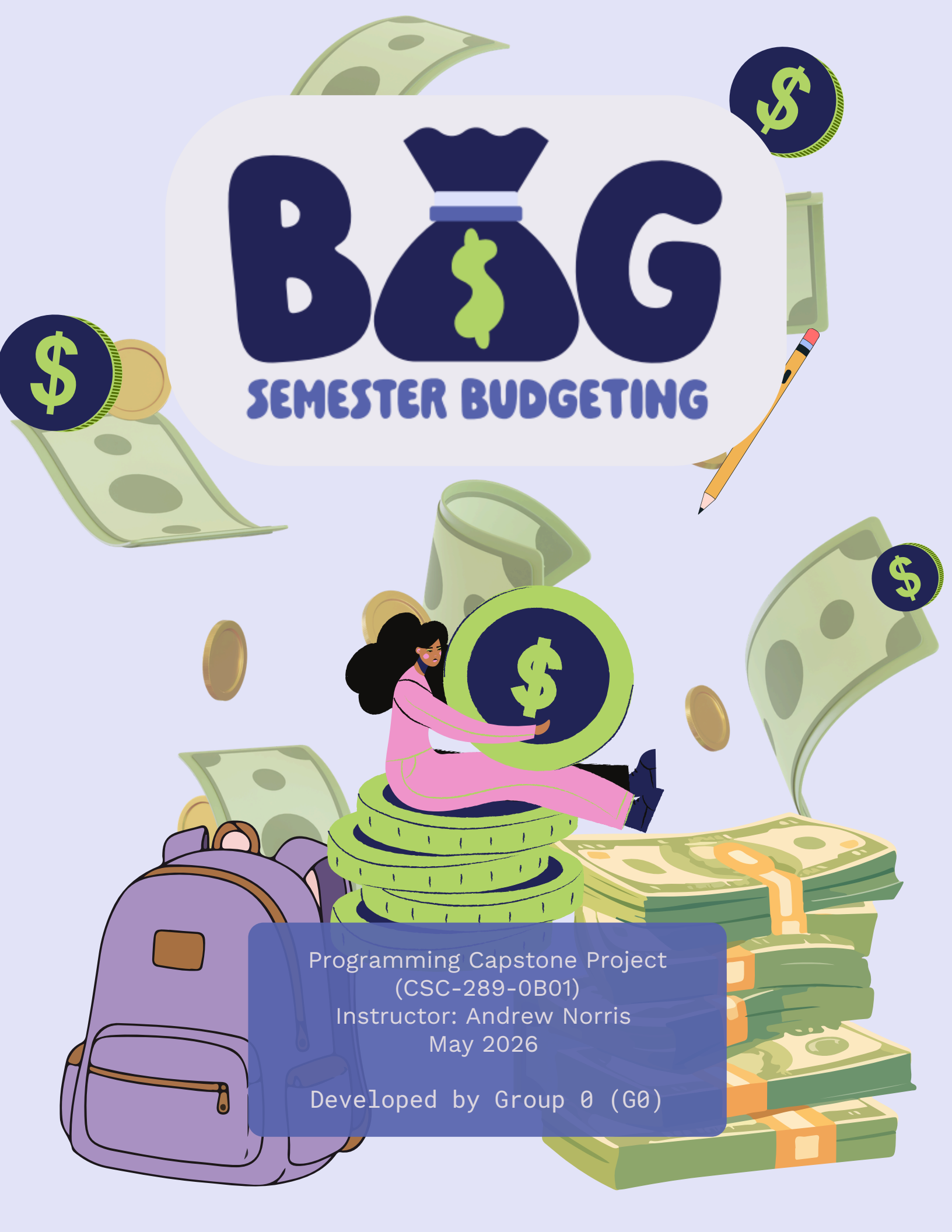


The background of the entire page is a light blue gradient. It is decorated with various floating dollar bills and coins. In the center, there is a large white rounded rectangle containing the title. Below the title, there is a blue semi-transparent box with white text. At the bottom left, there is a purple backpack. In the center, a person with long black hair, wearing a pink long-sleeved shirt and pink pants, is sitting on a stack of green coins. They are holding a large green coin with a white dollar sign. To the right of the person, there is a stack of yellow dollar bills. A yellow pencil is also visible, pointing towards the center.

BAG

SEMESTER BUDGETING

Programming Capstone Project
(CSC-289-0B01)
Instructor: Andrew Norris
May 2026

Developed by Group 0 (G0)

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PROJECT DESCRIPTION

BAG

Budgeting Assistance Guide

Semester Budgeting Application



BAG

(BUDGETING ASSISTANCE GUIDE)

SEMESTER BUDGETING APPLICATION

Project Overview

BAG is a semester-based budgeting application designed to help college students manage financial aid, track expenses, and monitor spending throughout an academic semester. Unlike traditional budgeting systems that focus on monthly income cycles, BAG organizes financial tracking around semester timelines and lump-sum funding distributions.

The application helps users compare spending activity against semester progression, allowing students to better pace expenses and maintain visibility into their remaining available funds.

Key Features

- Semester-based budgeting system
- Financial aid tracking and fund visibility
- Expense tracking with customizable categories
- Spending comparison against semester progress
- Budget goal management (weekly, monthly, or semester-based)
- Interactive dashboard with financial visualizations
- Secure account authentication and password recovery

Target Users

BAG was designed for college students who receive lump-sum financial aid payments such as FAFSA, scholarships, military benefits, or grant funding. The application supports students who need a structured way to organize finances and manage spending across an entire academic semester.



Take control of your semester finances.

BAG is a semester-based budgeting app designed for students managing financial aid, personal expenses, and school-related costs.

Track your funds, monitor spending by category, set budgeting goals, and see whether your money is lasting at the right pace for the semester.



Welcome

Create your account and start planning your semester with BAG.

Create Account

Register as a student or a parent/guardian.

Email

student@bag.com

Password

.....

Register as

Student

Security Question

What city were you born in?

Security Answer

Fayetteville

[Register](#)

Already have an account? [Log in](#)

Team & Product Planning

BAG

(Budgeting Assistance Guide)

Team Roles • Product Vision • User Stories



Team Roster & Roles

Overview

The BAG (Budgeting Assistance Guide) project was developed by Group 0 (G0) as part of the CSC-289 Programming Capstone course. Team members contributed to front-end development, back-end systems, database organization, interface design, debugging, and project documentation.

Team Members & Responsibilities

Joey Ackerman-Lowery

Front-end and back-end development, dashboard functionality, transaction management, portfolio design, and presentation materials.

Mattea Isley

User interface implementation, dashboard graphs, GitHub Pages organization, CSS modularization, and profile page development.

Paul Gayle

Back-end systems, server-side functionality, API integration, and password reset implementation.

Lydia Loffert

Database oversight, front-end improvements, Agile coordination, meeting documentation, and project organization.

Graphic Design Collaboration

Eva Daves

Graphic design, color palette development, branding direction, and visual identity concepts.

Product Vision

Overview

The vision for BAG (Budgeting Assistance Guide) was to develop a budgeting application specifically designed around the financial realities of college students. Unlike traditional budgeting tools that focus on monthly income cycles, BAG was designed to help students manage lump-sum financial aid across the duration of an academic semester.

Target Audience

- College students receiving financial aid
- Students managing scholarships or military benefits
- Users needing structured semester-based budgeting

Core Goals

- Help students avoid overspending early in the semester
- Provide clear visibility into available funds
- Compare spending against semester progression
- Create a simple and user-friendly budgeting experience

Long-Term Vision

Future versions of BAG could include mobile-responsive support, expanded financial analytics, and enhanced parent monitoring features for long-term budgeting assistance.

Why BAG Was Needed

Traditional budgeting applications focus on monthly income cycles and do not reflect how many college students receive financial aid. BAG addresses this gap by helping students manage lump-sum semester funding across an academic timeline.

User Stories

Student Budget Tracking

- As a student, I want to track my expenses so I can manage my money throughout the semester.
- As a student, I want to compare spending against semester progress so I can avoid overspending.

Financial Aid Management

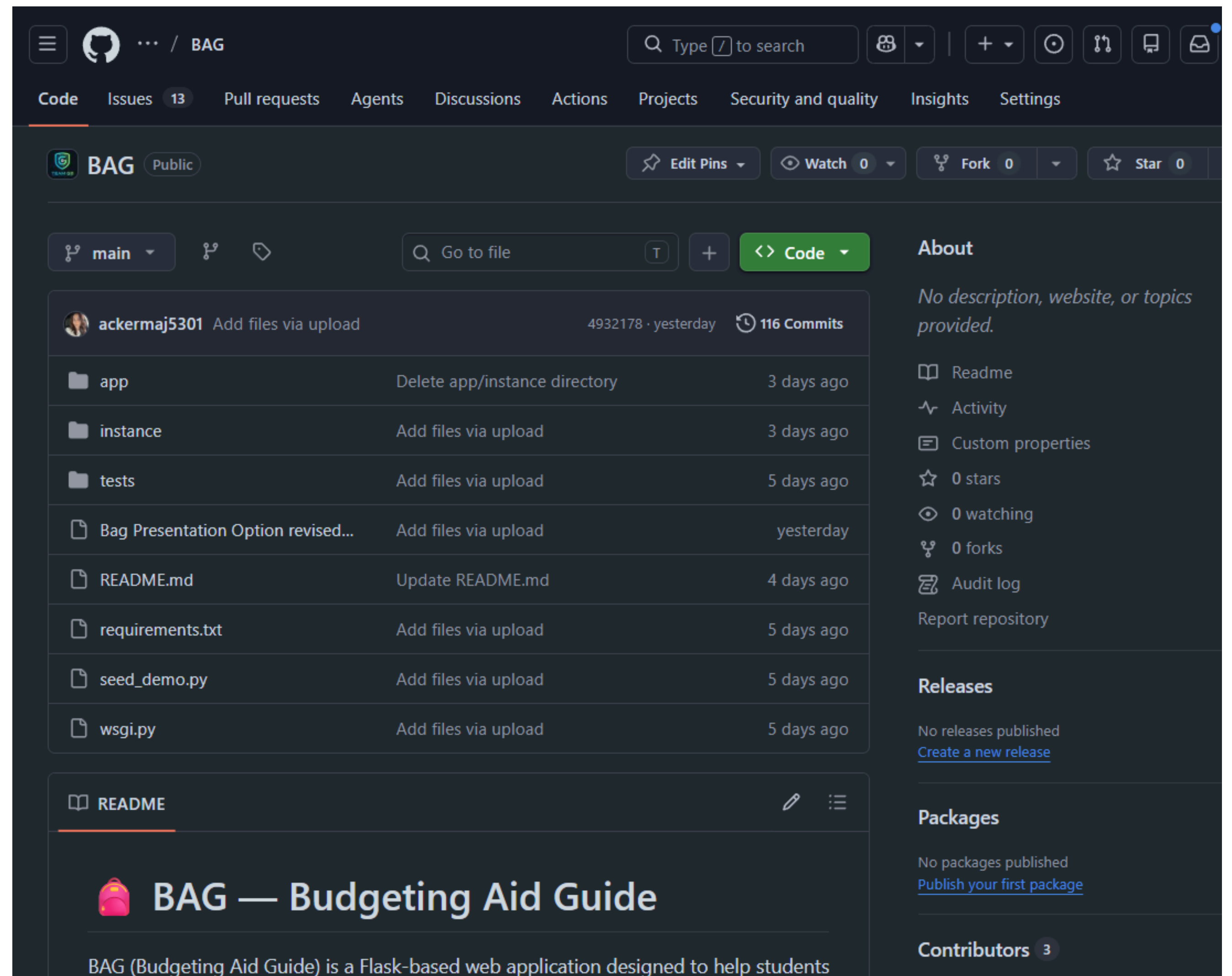
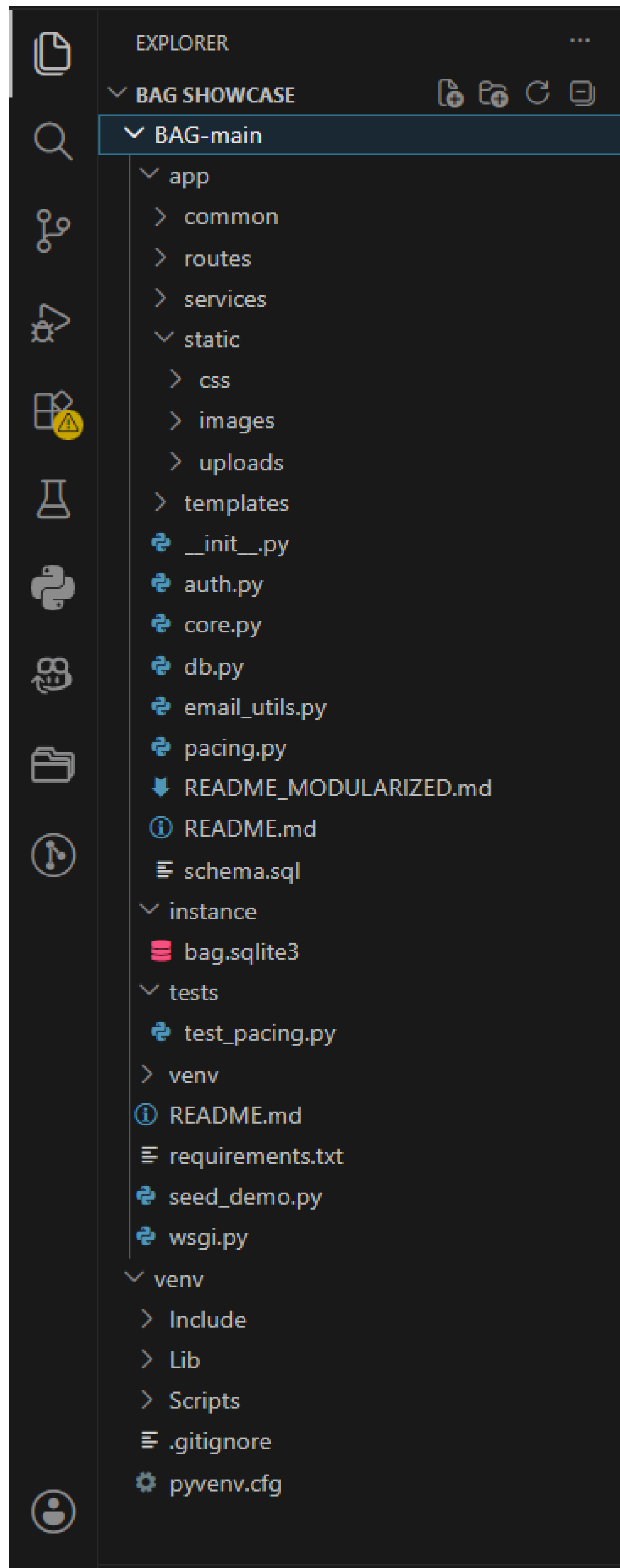
- As a student receiving financial aid, I want to input funding sources so I can monitor available semester funds.
- As a user, I want to organize aid by semester so I can better manage long-term finances.

Dashboard & Visualization

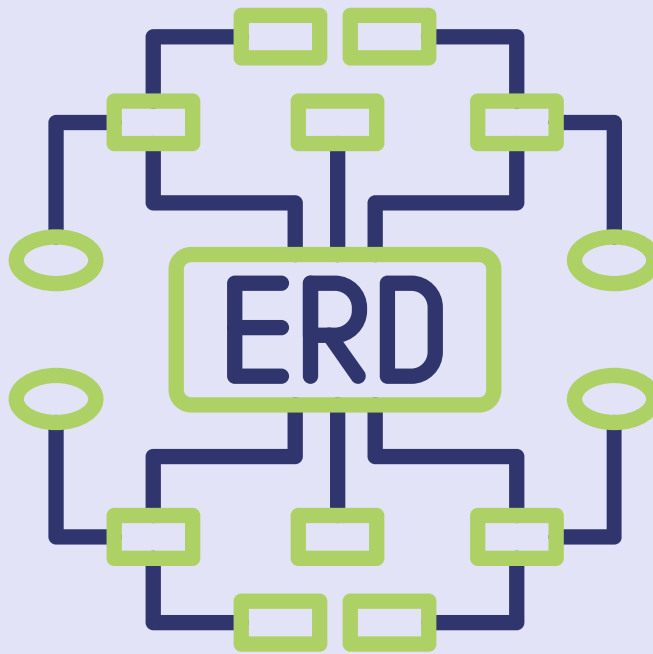
- As a user, I want visual spending graphs so I can quickly understand my financial status.
- As a student, I want easy access to budgeting information through a centralized dashboard.

Account & Security

- As a user, I want password reset functionality so I can securely recover my account.
- As a user, I want secure account authentication to protect my financial information.



Team G0



SYSTEM & DATABASE DESIGN

BAG

(Budgeting Assistance Guide)

**ERD • Data Dictionary • Key
Relationships**

System Architecture

Overview

BAG uses a full-stack web architecture that combines front-end user interaction, Flask-based server-side processing, and SQLite database management. The system was structured to support semester-based budgeting through modular application components and organized financial data storage.

Front-End Components

The front-end interface was developed using:

- HTML
- CSS
- JavaScript
- Jinja templates

These technologies support interactive pages, dashboard displays, forms, and navigation systems throughout the application.

Back-End Components

The back-end application was developed using:

- Python
- Flask

Flask manages route handling, server-side processing, authentication, and communication between the application interface and database system.

Database System

SQLite was used as the primary database management system. The database stores:

- User accounts
- Semester information
- Transactions
- Financial aid records
- Budget goals
- Categories

Database relationships allow financial records, budgeting data, and semester information to remain connected throughout the application workflow.

Development Tools

The development environment included:

- Visual Studio Code
- GitHub
- Git branching and version control
- ChartDB for ERD generation

These tools supported collaborative development, database planning, debugging, and project organization throughout the capstone process.

Application Workflow

Overview

BAG follows a structured workflow that connects user interaction, Flask route handling, and SQLite database storage. Users manage semesters, transactions, financial aid, and budgeting goals through a browser-based interface.

User Interaction Flow

1. Users create and manage academic semesters
2. Financial aid information is entered and tracked
3. Transactions are categorized and stored
4. Dashboard graphs update dynamically based on user activity
5. Budget progress is compared against semester progression

Data Processing

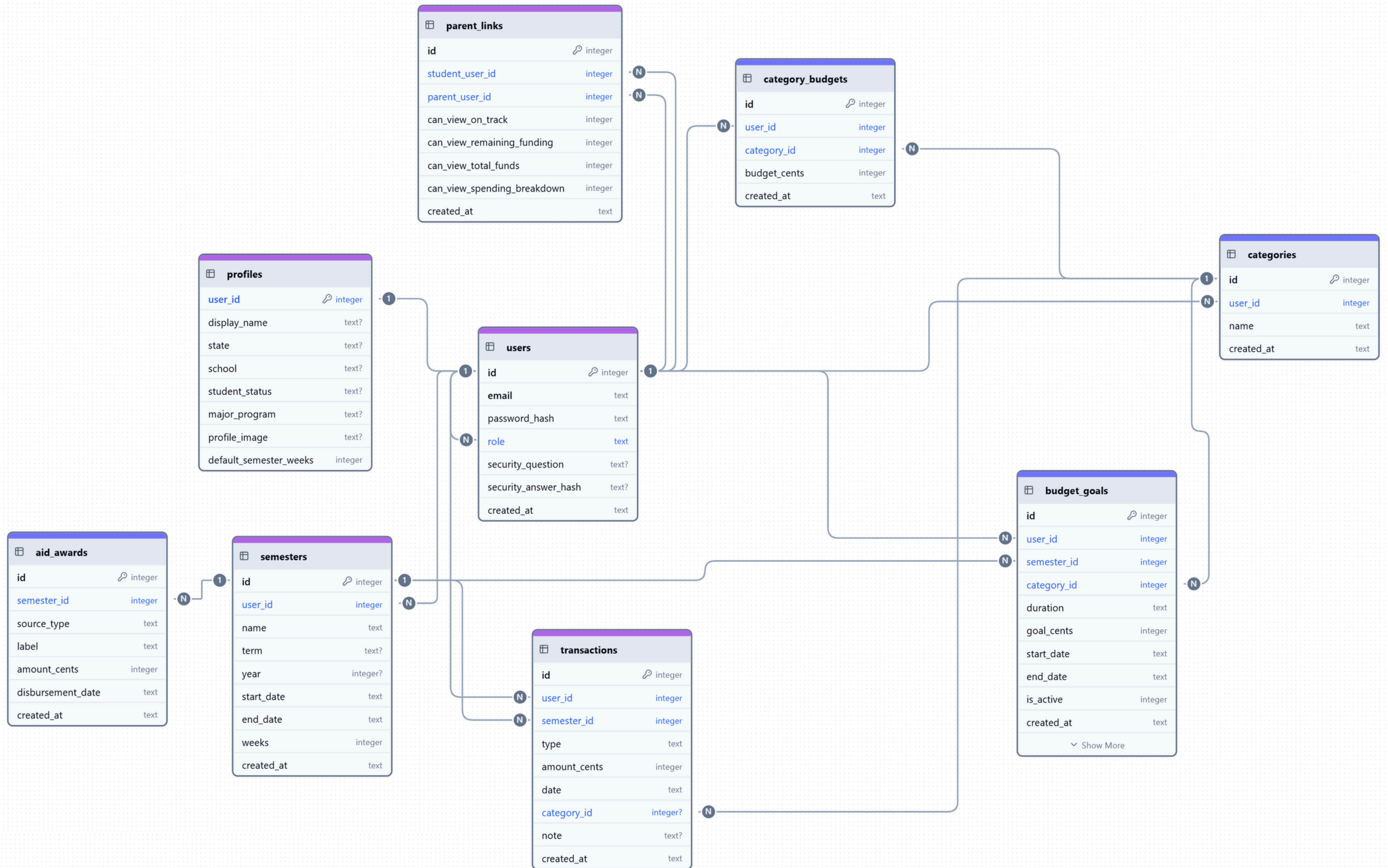
User input is processed through Flask route handling and validated before being stored in the SQLite database. Stored data is dynamically retrieved for dashboard updates, budgeting calculations, and financial tracking.

Database Storage

Semester information, financial aid records, transactions, categories, and budgeting goals are stored within the SQLite database. Relationships between tables allow financial data to remain organized and connected throughout the application workflow.

Dynamic Dashboard Updates

Dashboard displays automatically update based on changes to transactions, semester progress, and budgeting activity. This provides users with real-time insight into spending trends and remaining available funds.



Data Dictionaries

G Budget Goals Table

Key Columns

Column	Type	Constraint / Purpose
id	INTEGER	PK, auto-increment
user_id	INTEGER	FK → users.id
semester_id	INTEGER	FK → semesters.id
category_id	INTEGER	FK → categories.id
duration	TEXT	weekly, monthly, semester
goal_cents	INTEGER	Not null, > 0
is_active	INTEGER	Default 1

Key Relationship

- Links each active spending goal to a user, semester, and budget category.

BAG — Budgeting Assistance Guide

C Categories Table

Key Columns

Column	Type	Constraint / Purpose
id	INTEGER	PK, auto-increment
user_id	INTEGER	FK → users.id
name	TEXT	Not null
created_at	TEXT	Default timestamp
UNIQUE	user_id, name	Prevents duplicates

Key Relationship

- User-specific categories connect to transactions, category budgets, and budget goals.

U Users Table

Key Columns

Column	Type	Constraint / Purpose
id	INTEGER	PK, auto-increment
email	TEXT	Unique, not null
password_hash	TEXT	Not null
role	TEXT	student or parent
created_at	TEXT	Default timestamp

Key Relationship

- Referenced by profiles, semesters, categories, transactions, budget goals, and parent links.

T Transactions Table

Key Columns

Column	Type	Constraint / Purpose
id	INTEGER	PK, auto-increment
user_id	INTEGER	FK → users.id
semester_id	INTEGER	FK → semesters.id
type	TEXT	income or expense
amount_cents	INTEGER	Not null, ≥ 0
category_id	INTEGER	FK → categories.id
date	TEXT	Not null

Key Relationship

- Connects each income or expense transaction to a user, semester, and optional category.

BAG — Budgeting Assistance Guide

S Semesters Table

Key Columns

Column	Type	Constraint / Purpose
id	INTEGER	PK, auto-increment
user_id	INTEGER	FK → users.id
name	TEXT	Not null
term	TEXT	Optional
year	INTEGER	Optional
start_date	TEXT	Not null
end_date	TEXT	Not null
weeks	INTEGER	Not null

Key Relationship

- Defines the academic semester timeline used by aid awards, transactions, and budget goals.

BAG — Budgeting Assistance Guide

Key Relationships

User Relationships

Connected Tables:

- profiles
- semesters
- transactions
- categories
- budget_goals
- parent_links

Purpose:

Supports personalized budgeting data and account organization.

Semester Relationships

Connected Tables:

- transactions
- aid_awards
- budget_goals

Purpose:

Organizes budgeting activity around academic semesters.

Transaction Relationships

Connected Tables:

- users
- semesters
- categories

Purpose:

Tracks categorized financial activity across semester timelines.

Budget Goal Relationships

Connected Tables:

- users
- semesters
- categories

Purpose:

Supports category-based budgeting goals and spending analysis.

PROJECT CONTRIBUTIONS

BAG

(Budgeting Assistance Guide)



Development • Implementation • Collaboration

Joey Ackerman-Lowery

- Developed initial financial tracking application
- Established foundation for team development
- Enhanced Dashboard, Semesters, Transactions, and Funding sections
- Implemented edit and delete functionality
- Improved data organization and display
- Supported front-end and back-end integration
- Created visual portfolio and printed final copy
- Designed initial project pitch slideshow

GitHub Repository Access



Mattea Isley

- Developed and refined profile and budgeting pages
- Created intro screens to enhance user experience
- Built and organized GitHub Pages website
- Designed and implemented dashboard graphs
- Selected and integrated profile images and icons
- Implemented UI design and modularized CSS
- Contributed in remote development environment
- Assisted with troubleshooting and debugging

GitHub Repository Access



Paul Gayle

- Back-end development and system logic
- API integration and server-side functionality
- Implemented password reset features (email and security question)
- Assisted with core application systems
- Supported team collaboration and implementation
- Assisted with troubleshooting and debugging

GitHub Repository Access



Lydia Loffert

- Front-end design and user interface improvements
- Debugging and troubleshooting
- Oversight of database structure
- Coordinated team communication
- Guided Agile work practices
- Created portfolio layout document
- Maintained meeting notes
- Finalized presentation slideshow
- Assisted with showcase preparation
- Contributed to visual design direction

GitHub Repository Access



Eva Daves
Graphic Design Collaboration

- Assisted with the initial visual branding direction
- Helped develop the original BAG color palette and logo concepts
- Contributed early visual identity ideas for the project
- Supported the overall aesthetic direction of presentation materials

KEY FEATURES & IMPLEMENTATION

BAG

(Budgeting Assistance Guide)

Features • Functionality • Technical Notes



Key Features & Implementation

Overview

BAG includes core features that support semester-based budgeting, financial aid tracking, and expense management through an interactive financial dashboard. These tools help students better understand spending habits, remaining funds, and semester financial pacing.

Semester-Based Budgeting

- Tracks finances across an academic semester instead of a monthly cycle
- Compares spending to semester progress over time
- Helps users pace spending based on remaining funds

Financial Aid Tracking

- Allows users to input and manage multiple funding sources
- Calculates total available funds for each semester
- Integrates funding data into overall budget tracking

Transaction and Category Management

- Records expenses with date, amount, and category
- Allows users to edit and delete transactions
- Organizes spending using customizable categories

Interactive Dashboard

- Displays visual graphs of spending vs. time progression
- Updates dynamically based on user activity
- Provides quick insight into overall financial status

User Interface Design

- Clean and structured layout for ease of use
- Modular CSS supports consistent application styling
- Designed to simplify navigation and improve usability

Feature Implementation Examples

Dashboard Functionality

- Visual semester progress tracking
- Spending overview and budgeting insights
- Graphs comparing spending activity against semester progression
- Quick-access financial summary dashboard

Transaction Management

- Create, edit, and delete transaction records
- Category-based expense organization
- Semester-linked financial tracking
- Tracks both income and expense transactions

Semester-Based Budgeting

- Budgeting system organized around academic semesters
- Spending pace indicators based on semester progress
- Designed for lump-sum financial aid management
- Supports long-term semester expense planning

Account Security Features

- Secure user authentication system
- Password recovery and reset functionality
- Security question implementation
- Account management and accessibility support

Dashboard

On track

Hi Student One. Active semester: **Spring 2026**

Week

16 / 16

Funds left

\$3501.05

Semester elapsed

100.0%

Funds spent

33.9%

Nice work — you’re spending below pace for week 16. That gives you more flexibility later in the semester.

Totals

Total funds	Spent	Remaining
\$5300.00	\$1798.95	\$3501.05

Projection: At the current average pace, you may run out around week 46.

+ Add Funds

+ Add Expense

Semester elapsed

100.0%

Funds spent

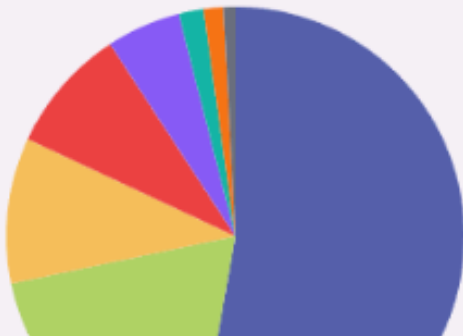
SAFE ZONE 33.9%

Income and Funding

Date	Source	Label	Amount	Actions
2026-05-04	Income	Paycheck	+\$1200.00	EditDelete
2026-02-20	Other	Birthday Gift	+\$500.00	EditDelete
2026-01-15	FAFSA	Spring financial aid refund	+\$3600.00	EditDelete

Spending by category

Category	Total
Housing	\$950.00
Food	\$339.45
Textbooks	\$185.00
Personal	\$159.00
Transportation	\$94.00

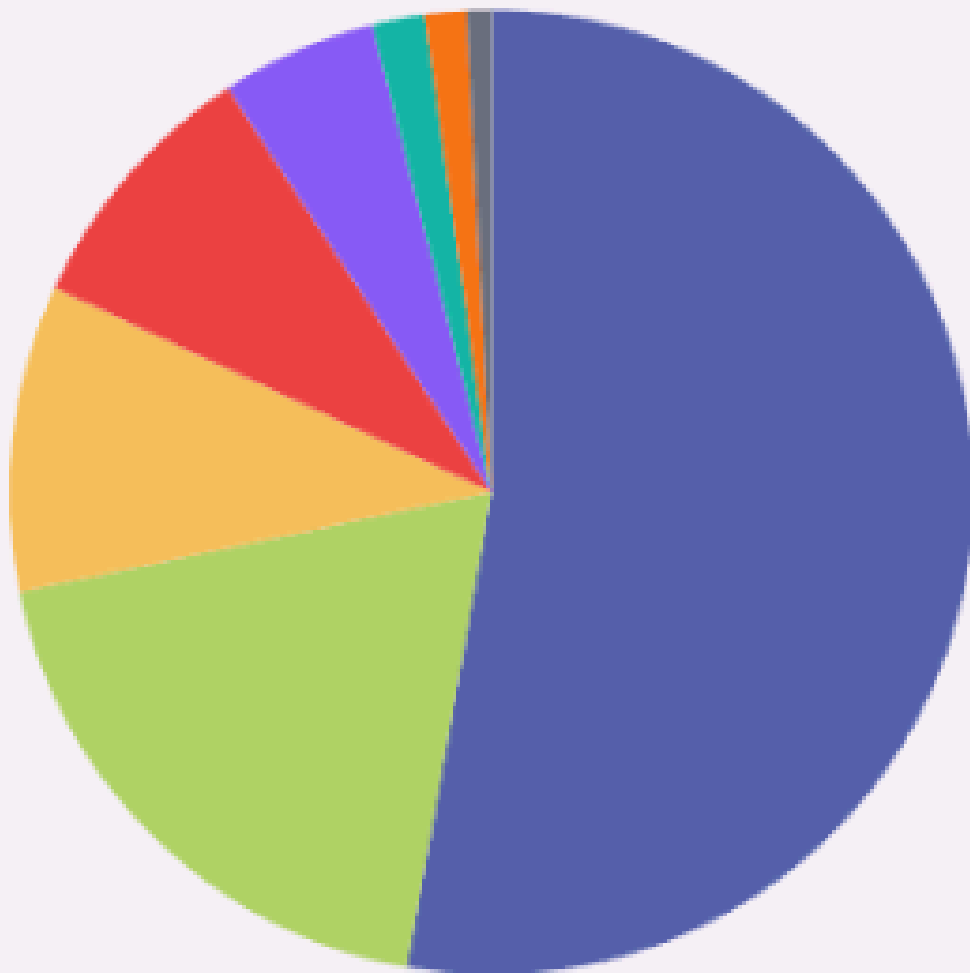


Recent transactions

Date	Category	Amount	Note	Actions
2026-05-03	Personal	-\$120.00	Hygiene	EditDelete
2026-05-03	Other	-\$15.00	Candles	EditDelete
2026-05-03	Food	-\$20.00	Doordash	EditDelete
2026-05-03	Food	-\$100.00	Dinner with friends	EditDelete

Spending by category

Category	Total
Housing	\$950.00
Food	\$339.45
Textbooks	\$185.00
Personal	\$159.00
Transportation	\$94.00
School Supplies	\$31.50
Health	\$25.00
Other	\$15.00

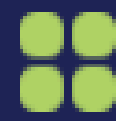


Recent transactions

Date	Category	Amount	Note	Actions	
2026-05-03	Personal	-\$120.00	Hygiene	Edit	Delete
2026-05-03	Other	-\$15.00	Candles	Edit	Delete
2026-05-03	Food	-\$20.00	Doordash	Edit	Delete
2026-05-03	Food	-\$100.00	Dinner with friends	Edit	Delete
2026-05-01	Transportation	-\$52.00	Gas	Edit	Delete
2026-05-01	Housing	-\$950.00	Rent	Edit	Delete
2026-04-24	Food	-\$16.75	Lunch on campus	Edit	Delete
2026-02-14	Food	-\$74.20	Groceries	Edit	Delete
2026-02-10	School Supplies	-\$31.50	Notebook and pens	Edit	Delete
2026-02-10	Health	-\$25.00	Pharmacy	Edit	Delete



SECURE YOUR BAG.

 Dashboard

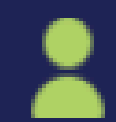
 Semesters

 Add Funds

 Add Expense

 Statements

 Budgeting

 Profile

 Parent Access

 FAQs

 Logout

Student One



Statements

Full transaction history for your account.

Date	Semester	Category	Type	Amount	Note	Actions	
2026-05-04	Spring 2026	Income	Income	+ \$1200.00	Paycheck	Edit	Delete
2026-05-03	Spring 2026	Food	Expense	- \$100.00	Dinner with friends	Edit	Delete
2026-05-03	Spring 2026	Food	Expense	- \$20.00	Doordash	Edit	Delete
2026-05-03	Spring 2026	Other	Expense	- \$15.00	Candles	Edit	Delete
2026-05-03	Spring 2026	Personal	Expense	- \$120.00	Hygiene	Edit	Delete
2026-05-03	Spring 2026	Other	Income	+ \$500.00	Birthday Gift	Edit	Delete
2026-05-01	Spring 2026	Housing	Expense	- \$950.00	Rent	Edit	Delete



SECURE YOUR BAG.

Dashboard

Semesters

Add Funds

Add Expense

Statements

Budgeting

Profile

Parent Access

FAQs

Logout

Student One



Budgeting

Set optional category goals and track spending from your expenses.

Active semester: **Spring 2026** (2026-01-14 to 2026-05-13)

[+ Add Goal](#)

Weekly Goals



SAFE ZONE

Other

Weekly spending goal

\$75.00

Allocated

\$15.00

Spent



\$60.00 left

20% used

[Edit](#)

[Delete](#)



WARNING

Transportation

Weekly spending goal

\$100.00

Allocated

\$52.00

Spent



\$48.00 left

52% used

[Edit](#)

[Delete](#)



DANGER

Food

Weekly spending goal

\$150.00

Allocated

\$120.00

Spent



\$30.00 left

80% used

[Edit](#)

[Delete](#)

Monthly Goals



WARNING

Housing

Monthly spending goal

\$1800.00

Allocated

\$950.00

Spent



\$850.00 left

53% used

[Edit](#)

[Delete](#)



DANGER

Personal

Monthly spending goal

\$180.00

Allocated

\$159.00

Spent



\$21.00 left

88% used

[Edit](#)

[Delete](#)



SAFE ZONE

Transportation

Monthly spending goal

\$220.00

Allocated

\$94.00

Spent



\$126.00 left

43% used

[Edit](#)

[Delete](#)



SAFE ZONE

Food

Monthly spending goal

\$600.00

Allocated

\$120.00

Spent



\$480.00 left

20% used

[Edit](#)

[Delete](#)

Semester Goals



SAFE ZONE

Textbooks

Semester spending goal

\$500.00

Allocated

\$185.00

Spent



\$315.00 left

37% used

[Edit](#)

[Delete](#)



SECURE YOUR BAG.

 Dashboard

 Semesters

 Add Funds

 Add Expense

 Statements

 Budgeting

 Profile

 Parent Access

 FAQs

 Logout

Student One



Add Funds

Source type

Select a source type

Select a source type

Income

FAFSA

GI Bill

Scholarship

Loan

Savings

Other

name, etc

Date

05/06/2026



```
7
8 <div class="card">
9   <h2>Add Funds </h2>
10  <form method="post" class="form grid">
11    <div>
12      <label>Source type</label>
13      <select name="source_type">
14        <option value="" disabled selected>Select a source type</option>
15        <option>Income</option>
16        <option>FAFSA</option>
17        <option>GI Bill</option>
18        <option>Scholarship</option>
19        <option>Loan</option>
20        <option>Savings</option>
21        <option>Other</option>
22      </select>
23    </div>
24    <div class="full">
25      <label>Label</label>
26      <input name="label" placeholder="FAFSA refund, GI Bill housing, Scholarship name, etc">
27    </div>
28    <div>
29      <label>Amount ($)</label>
30      <input type="number" step="0.01" min="0.01" name="amount" required>
31    </div>
32    <div>
33      <label>Date</label>
34      <input type="date" name="disbursement_date" required>
35    </div>
36    <div class="full">
37      <button class="btn" type="submit">Save Funds</button>
38    </div>
39  </form>
40 </div>
41 {% endblock %}
```



SECURE YOUR BAG.

 Dashboard

 Semesters

 Add Funds

 Add Expense

 Statements

 Budgeting

 Profile

 Parent Access

 FAQs

 Logout

Student One



Add Expense

Category

Select category

Select category

Food

Health

Housing

Other

Personal

School Supplies

Textbooks

Transportation

Date

05/06/2026



```
11 <form method="post" class="form grid">
12   <input type="hidden" name="type" value="expense">
13
14   <div>
15     <label>Category</label>
16     <select name="category_id" required>
17       <option value="" disabled selected>Select category</option>
18       {% for c in categories %}
19         <option value="{{ c.id }}">{{ c.name }}</option>
20       {% endfor %}
21     </select>
22   </div>
23
24   <div class="full">
25     <label>Note</label>
26     <input name="note" placeholder="Shopping, etc">
27   </div>
```


Forgot Password

Enter your email and we'll send you a password reset link.

Email

Send Reset Link

[Reset password with security question](#)

Remember your password? [Back to login](#)

Security Question Reset

Enter your email to find your security question.

Email

Next

[Back to login](#)

Answer Security Question

What city were you born in?

Your Answer

Submit Answer

[Back to login](#)

DEPLOYMENT



Environment • Technologies • Execution

BAG
(Budgeting Assistance Guide)

Deployment

Overview

BAG was deployed as a locally hosted Flask web application that allows users to manage semester-based budgeting, financial aid tracking, and expense organization through a browser-based interface.

Technologies Used

- Python
- Flask
- SQLite
- HTML
- CSS
- JavaScript

Development Environment

The project was developed using Visual Studio Code and GitHub for collaboration and version control. Team members used Git branches and repositories to manage project updates and merge development changes.

Running the Application

To run the application locally:

1. Install project dependencies
2. Initialize the database
3. Run the Flask application
4. Open the local host address in a web browser

Future Deployment Considerations

Future deployment improvements could include cloud hosting, mobile-responsive support, and expanded parent monitoring accessibility for broader student use.

Dependencies

Dependencies were managed through a requirements.txt file containing Flask, SQLite integration libraries, and authentication support packages.


```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS flask - BAG-main + - - - - -
```

```
PS C:\Users\joeyn\Documents\CSC-289\Bag ShowCase> cd BAG-main
PS C:\Users\joeyn\Documents\CSC-289\Bag ShowCase\BAG-main> python -m venv venv
PS C:\Users\joeyn\Documents\CSC-289\Bag ShowCase\BAG-main> venv\Scripts\activate
(venv) PS C:\Users\joeyn\Documents\CSC-289\Bag ShowCase\BAG-main> pip install -r requirements.txt
Requirement already satisfied: Flask==3.0.3 in .\venv\Lib\site-packages (from -r requirements.txt (line 1)) (3.0.3)
Requirement already satisfied: Werkzeug==3.0.3 in .\venv\Lib\site-packages (from -r requirements.txt (line 2)) (3.0.3)
Requirement already satisfied: pytest==8.3.2 in .\venv\Lib\site-packages (from -r requirements.txt (line 3)) (8.3.2)
Requirement already satisfied: itsdangerous==2.2.0 in .\venv\Lib\site-packages (from -r requirements.txt (line 4)) (2.2.0)
Requirement already satisfied: dotenv==0.9.9 in .\venv\Lib\site-packages (from -r requirements.txt (line 5)) (0.9.9)
Requirement already satisfied: Jinja2>=3.1.2 in .\venv\Lib\site-packages (from Flask==3.0.3->-r requirements.txt (line 1)) (3.1.6)
Requirement already satisfied: click>=8.1.3 in .\venv\Lib\site-packages (from Flask==3.0.3->-r requirements.txt (line 1)) (8.3.3)
Requirement already satisfied: blinker>=1.6.2 in .\venv\Lib\site-packages (from Flask==3.0.3->-r requirements.txt (line 1)) (1.9.0)
Requirement already satisfied: MarkupSafe>=2.1.1 in .\venv\Lib\site-packages (from Werkzeug==3.0.3->-r requirements.txt (line 2)) (3.0.3)
Requirement already satisfied: iniconfig in .\venv\Lib\site-packages (from pytest==8.3.2->-r requirements.txt (line 3)) (2.3.0)
Requirement already satisfied: packaging in .\venv\Lib\site-packages (from pytest==8.3.2->-r requirements.txt (line 3)) (26.2)
Requirement already satisfied: pluggy<2,>=1.5 in .\venv\Lib\site-packages (from pytest==8.3.2->-r requirements.txt (line 3)) (1.6.0)
Requirement already satisfied: colorama in .\venv\Lib\site-packages (from pytest==8.3.2->-r requirements.txt (line 3)) (0.4.6)
Requirement already satisfied: python-dotenv in .\venv\Lib\site-packages (from dotenv==0.9.9->-r requirements.txt (line 5)) (1.2.2)

[notice] A new release of pip is available: 26.0.1 -> 26.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS C:\Users\joeyn\Documents\CSC-289\Bag ShowCase\BAG-main> flask run
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [06/May/2026 17:55:08] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [06/May/2026 17:55:08] "GET /static/images/logos/baglogo.png HTTP/1.1" 304 -
127.0.0.1 - - [06/May/2026 17:55:08] "GET /static/images/logos/baglogotransparent.png HTTP/1.1" 304 -
127.0.0.1 - - [06/May/2026 17:55:08] "GET /static/css/introstyle.css HTTP/1.1" 304 -
127.0.0.1 - - [06/May/2026 17:56:48] "GET /login HTTP/1.1" 200 -
```

```
≡ requirements.txt X
```

```
BAG-main > ≡ requirements.txt
```

```
1  Flask==3.0.3
2  Werkzeug==3.0.3
3  pytest==8.3.2
4  itsdangerous==2.2.0
5  dotenv==0.9.9
```

Reflection & Lessons Learned

BAG
(BUDGETING ASSISTANCE GUIDE)
GROWTH • CHALLENGES • OUTCOMES



Reflection & Lessons Learned

Overview

Developing the BAG (Budgeting Assistance Guide) application provided valuable experience in both technical development and collaborative software design. Throughout the project, the team applied programming, database management, and user interface design principles to create a functional semester-based budgeting system.

Technical Growth

Through this project, the team gained experience with:

- Front-end and back-end application development
- Flask route handling and server-side processing
- SQLite database organization and management
- Dashboard functionality and financial data visualization
- Debugging, troubleshooting, and feature integration

The project strengthened our understanding of how application components work together within a full-stack system.

Collaboration & Teamwork

The development process required:

- Consistent communication between team members
- Coordination across multiple project components
- GitHub collaboration and version control management
- Shared problem-solving and debugging efforts
- Adaptability throughout the development process

Working collaboratively reinforced the importance of organization, accountability, and effective teamwork in software development.

Challenges Encountered

Some of the primary challenges included:

- Integrating features developed by multiple team members
- Maintaining consistency across front-end and back-end systems
- Troubleshooting routing, database, and interface issues
- Managing project updates and development timelines

Addressing these challenges improved both technical troubleshooting skills and overall project coordination.

Reflection & Lessons Learned

Project Outcomes

The final application successfully combines:

- Semester-based budgeting
- Financial aid tracking
- Expense and transaction management
- Interactive dashboard functionality
- Organized financial data storage

The completed system demonstrates both technical development skills and the ability to collaboratively build a functional full-stack application.

Future Improvements

Future development could include:

- Expanded customization options
- Enhanced dashboard visualizations
- Mobile-responsive optimization
- Advanced financial reporting tools
- Additional accessibility and usability improvements

These additions could further improve long-term budgeting support and overall user experience.

Bag

Secure your bag. Own your independence.

Dashboard

Add Transaction

Add Aid

Semesters

Categories

Profile

Logout

Semester created and selected.

Dashboard

On track

Active semester: My Semester

Week

7 / 16

Safe to spend this week

\$0.00

Semester elapsed

43.8%

Funds spent

0.0%

Nice work — you're spending below pace for week 7. That gives you more flexibility later in the semester.

Totals

Total funds

\$0.00

Spent

\$0.00

Remaining

\$0.00

Projection will appear after you add some expenses.

+ Add Aid

+ Add Transaction

Spending by category

No categorized expenses yet.

Recent transactions

No transactions yet.

Bag

Secure your bag. Own your independence.

Login

Register

Login

Email

Password

Login

No account? Register

Bag capstone prototype · Flask + SQLite

Bag

Secure your bag. Own your independence.

Dashboard

Add Transaction

Add Aid

Semesters

Categories

Profile

Logout

Categories

Add a category (e.g., Dining Out)

Add

Food

Health

Housing

Other

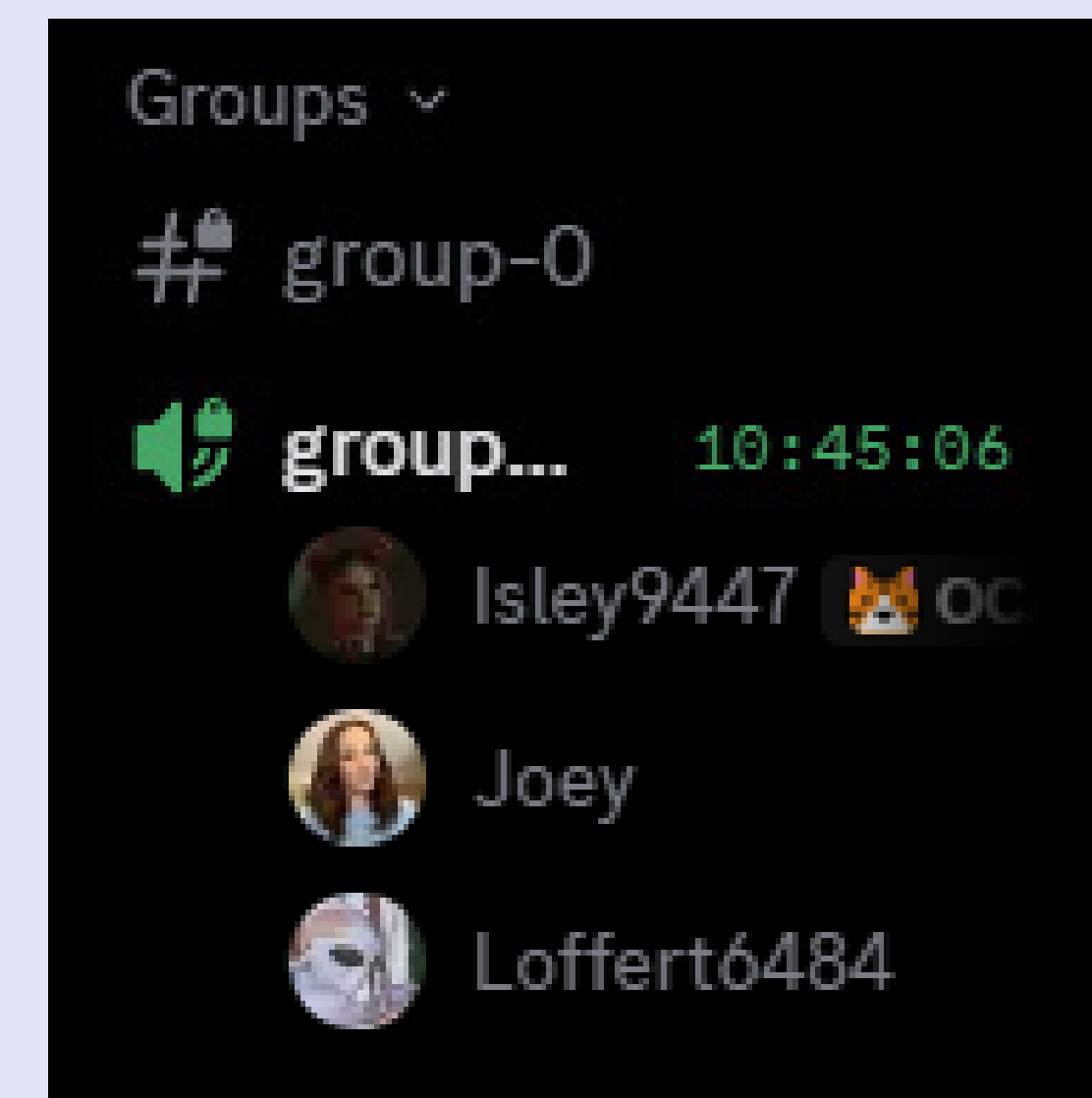
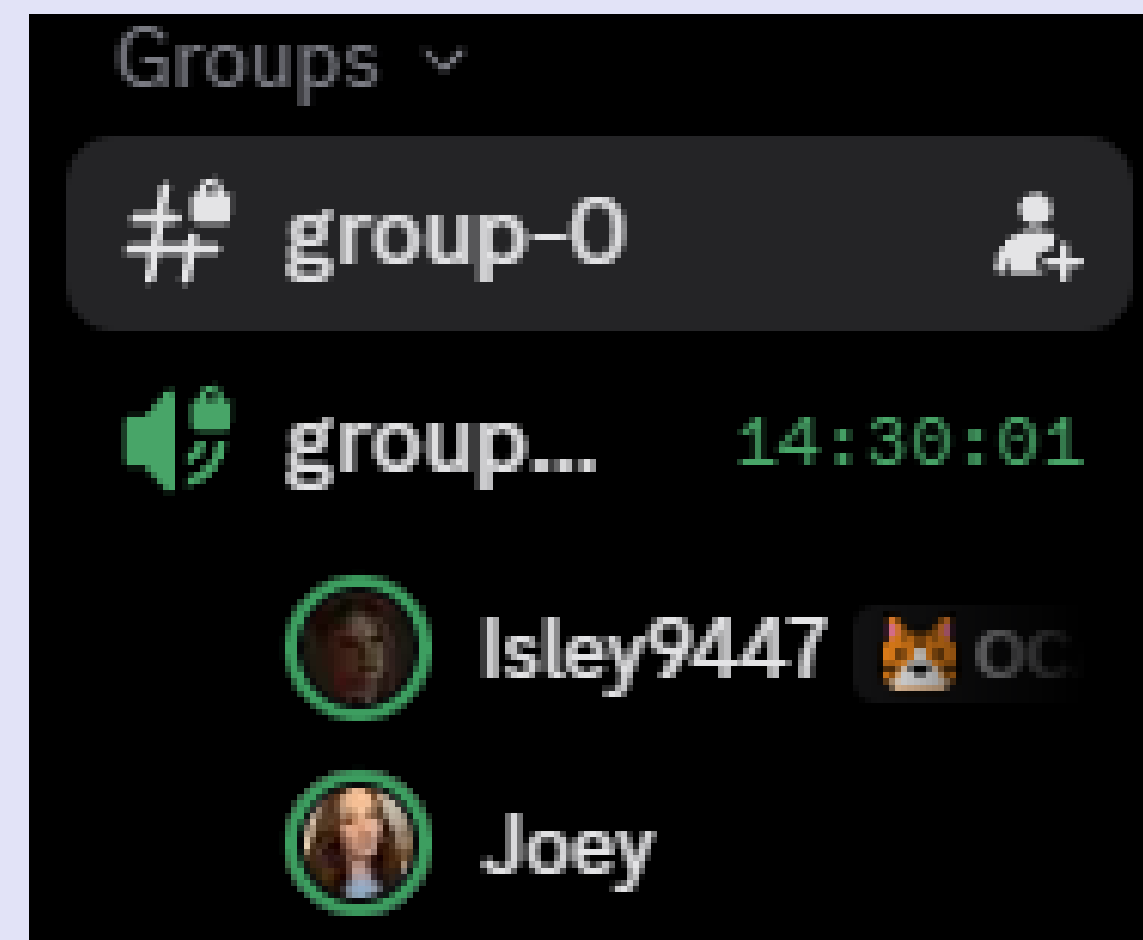
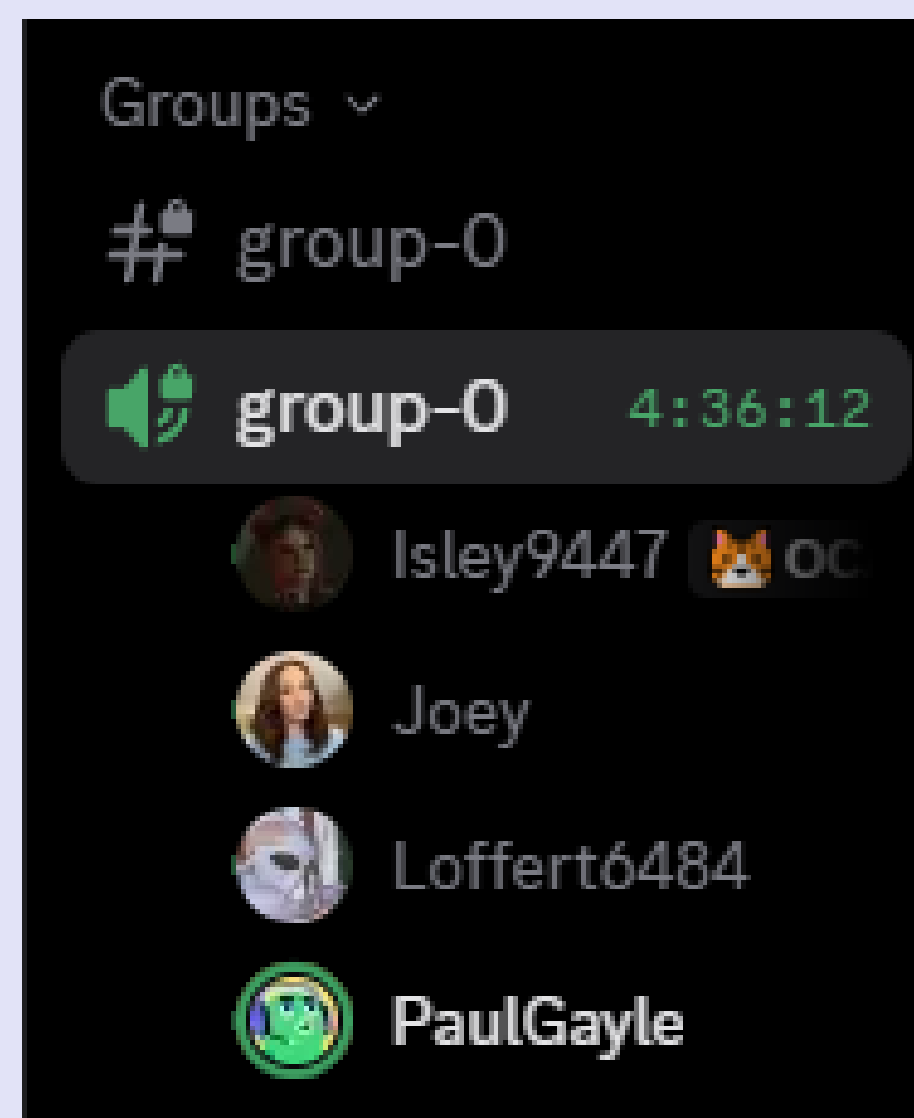
Personal

School Supplies

Textbooks

Transportation

Bag capstone prototype · Flask + SQLite



3/30/2026 6:00PM Meeting

LOOSE AGENDA:

- * Review Joey's parental view PR
- * Demo Paul's password reset feature
- * Loffert show Google Stitch BAG wireframe
- * Review Mattea FAQ and profile page

LOOSE MINUTES:

Loffert showed group google stitch output and html code viewer

Joey demo-ed Parent view and parent access, including new "show password" feature when logging/signing in.

****Discussed adding ability to edit linked parent accounts, so you don't have to re-type parent email**

Mattea showcase profile page and FAQ page

Paul showcased password reset feature. Works!

****Discussed adding security questions as a alternative to email verification**

Merged Joey parent view changes to main

Discussed and got confused about how GitHub merging works again. Merged mattea-testbranch into main

3/17/2026 6:00PM Meeting

LOOSE AGENDA:

*Paul check-in (password reset feature)

*Joey check-in

*Mattea check-in

*look at issues board

**Go thru using app (test each function/simulate an everyday use of Bag)*

***Have something to put on GitHub by end of meeting (file or issue update etc, for GitHub activity log)**

stretch goals brought up:

*what to expect page (general academic costs to expect)

*alert if "part-time student" selected that you might not received/be eligible for benefits

*bag upload profile pictures

*gamification elements (stickers collectables)

Copyright & Ownership

Portfolio Design & Documentation

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Project materials, collaborative software components, and shared BAG application assets are associated with Group_0.

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